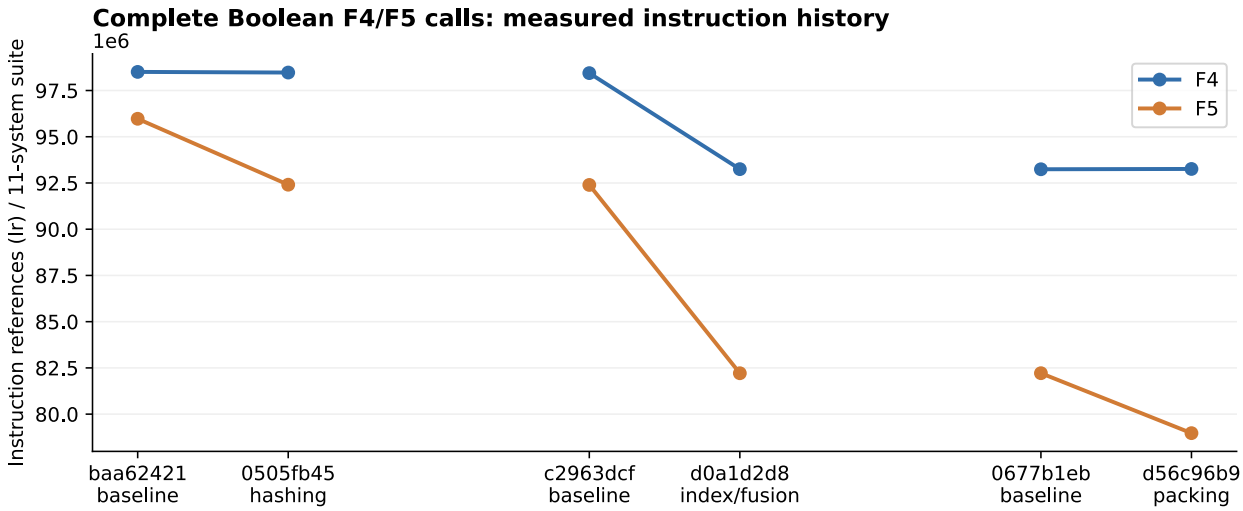
Complete-call instruction history



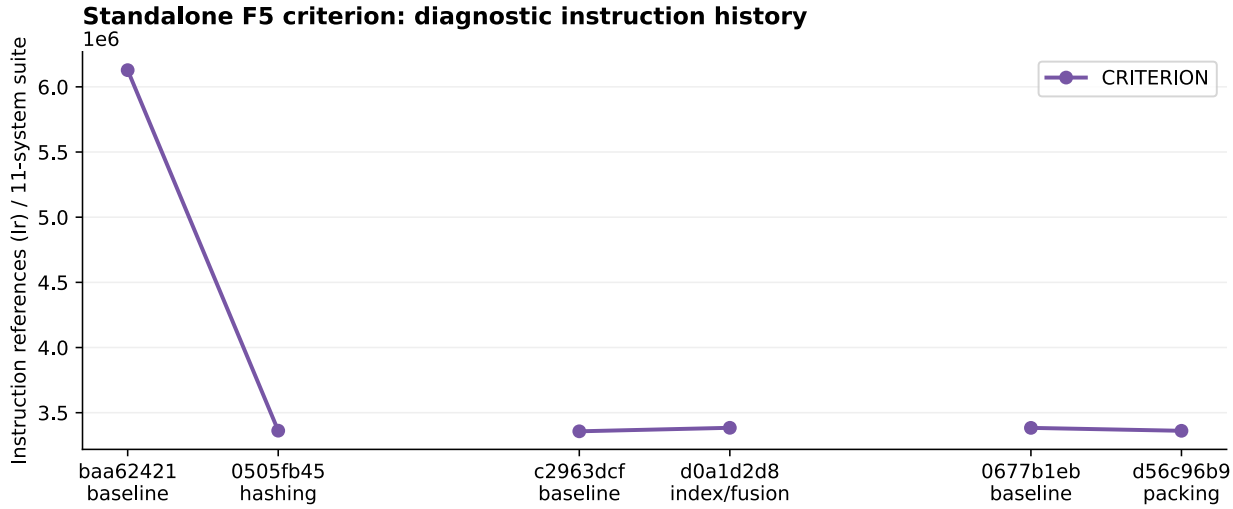
Three independently matched episodes; gaps do not imply cumulative gains.
Each endpoint: 5 native paired samples and median; complete merged-worker dumps. run-20261012T1503Z

F4: 93,241,594 to 93,258,282 Ir; paired reduction −0.0199% (95% CI −0.0258 to −0.0109%); neutral. F5: 82,213,799 to 78,971,828 Ir; paired reduction 3.9472% (95% CI 3.9239 to 3.9574%); fewer instructions. CRITERION: 3,383,582 to 3,361,188 Ir; paired reduction 0.6618% (95% CI 0.6618 to 0.6754%); neutral. Preregistered F5/complete-call guard: True. Memory regression entries: 0. All case verdicts, including criterion regressions, remain in the native comparison.

Local validation: Release library: 3,996 passed, 0 failed, 112 ignored, exit 0, 238.05 s with loopback permission. Initial restricted run: 3,987 passed, 9 socket-permission failures, 112 ignored; retained. Focused suites: 47 Boolean Groebner, 18 F5, 21 PQ Groebner; 37 loopback checks; comparator 5 tests. All pass. Cargo example targets: 5 and 0 tests, both pass. Strict Clippy on both examples passes. Runtime ELF code/data equivalence passes.

run-20261012T1503Z; baseline 0677b1eb; candidate d56c96b9; AMD EPYC 9V74 Linux x86-64; Rust 1.99.0; Callgrind 3.24; one Rayon worker; 18 processes / 792 audited calls plus the 33-call hash control. No elapsed-time, process RSS or full-solver gain measured. Sources: native results and raw hashes, frozen protocol, report, local validation.

Criterion instruction diagnostics



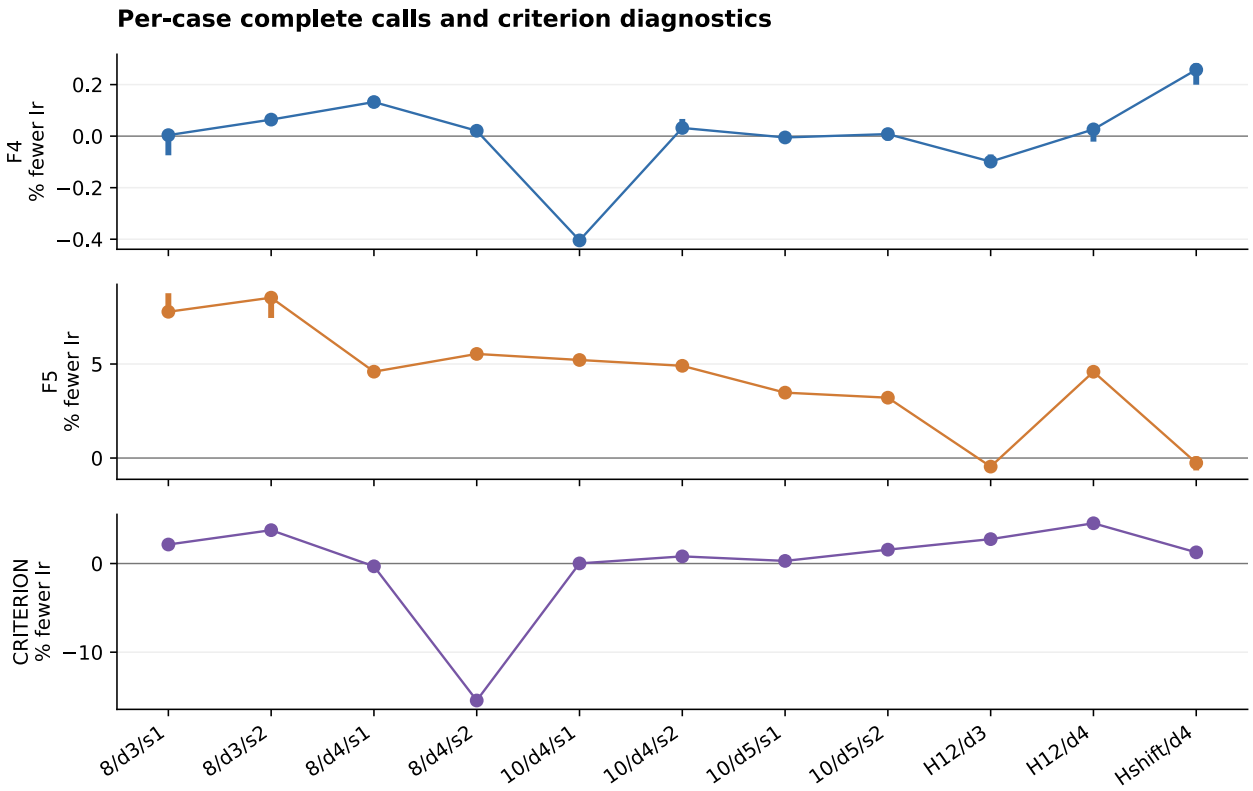
Three independently matched episodes; gaps do not imply cumulative gains.
Each endpoint: 5 native paired samples and median; complete merged-worker dumps. run-20261012T1503Z

Criterion construction is measured separately from complete calls. One n8/d4/s2 case records 15.4494% more instructions; the cause is not established. Unchanged source does not establish unchanged generated instruction costs. F4: 93,241,594 to 93,258,282 Ir; paired reduction −0.0199% (95% CI −0.0258 to −0.0109%); neutral. F5: 82,213,799 to 78,971,828 Ir; paired reduction 3.9472% (95% CI 3.9239 to 3.9574%); fewer instructions. CRITERION: 3,383,582 to 3,361,188 Ir; paired reduction 0.6618% (95% CI 0.6618 to 0.6754%); neutral.

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Per-case results and uncertainty



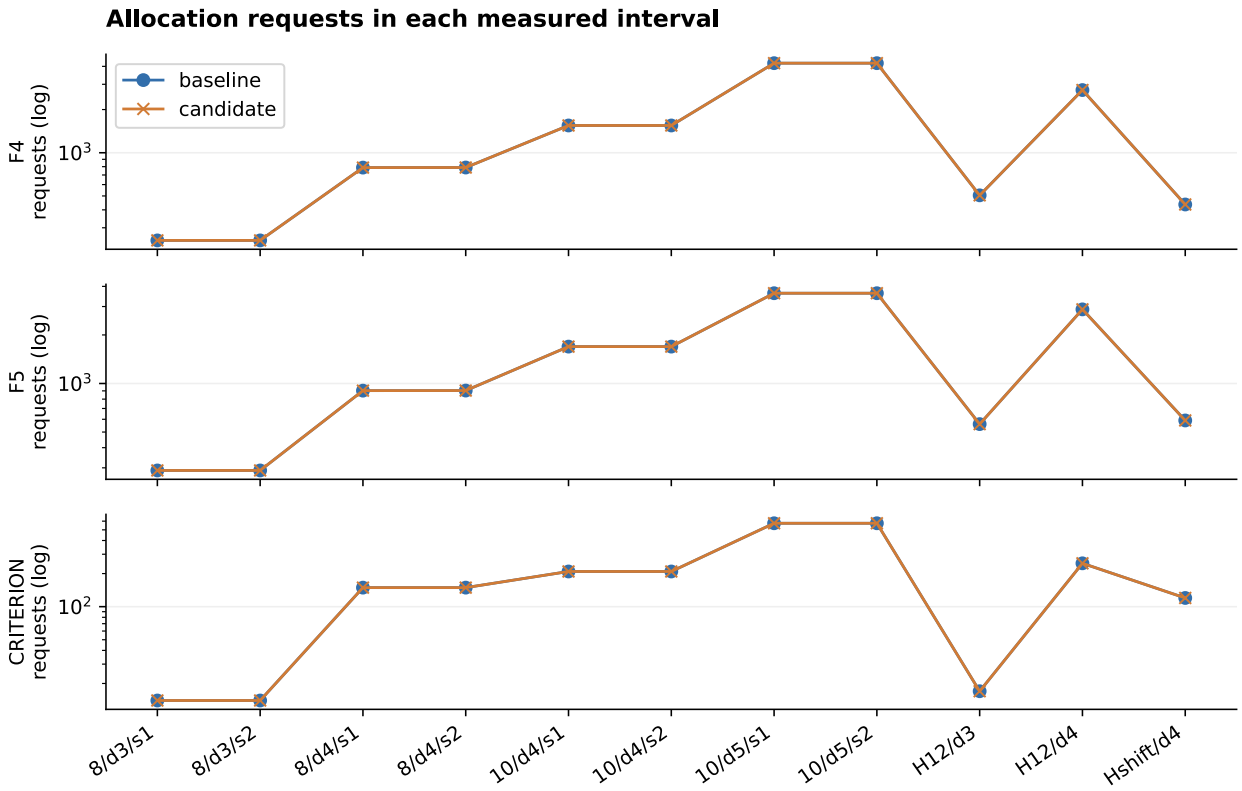
5 paired rounds; native reduction estimates and 95% bootstrap intervals. Positive = fewer Ir.
H = known frozen regression fixture; Hshift uses bits 52–63 and 6 multiplier variables.
run-20261012T1503Z · Source: records/comparison.json.

All eleven cases and all three arms are shown, including neutral and more-instruction verdicts. The frozen gate requires no complete-call regression, with standalone criterion Ir reported separately. Preregistered F5/ complete-call guard: True. Memory regression entries: 0. All case verdicts, including criterion regressions, remain in the native comparison.

Local validation: Release library: 3,996 passed, 0 failed, 112 ignored, exit 0, 238.05 s with loopback permission. Initial restricted run: 3,987 passed, 9 socket-permission failures, 112 ignored; retained. Focused suites: 47 Boolean Groebner, 18 F5, 21 PQ Groebner; 37 loopback checks; comparator 5 tests. All pass. Cargo example targets: 5 and 0 tests, both pass. Strict Clippy on both examples passes. Runtime ELF code/data equivalence passes.

run-20261012T1503Z; baseline 0677b1eb; candidate d56c96b9; AMD EPYC 9V74 Linux x86-64; Rust 1.99.0; Callgrind 3.24; one Rayon worker; 18 processes / 792 audited calls plus the 33-call hash control. No elapsed-time, process RSS or full-solver gain measured. Sources: native results and raw hashes, frozen protocol, report, local validation.

Allocation requests



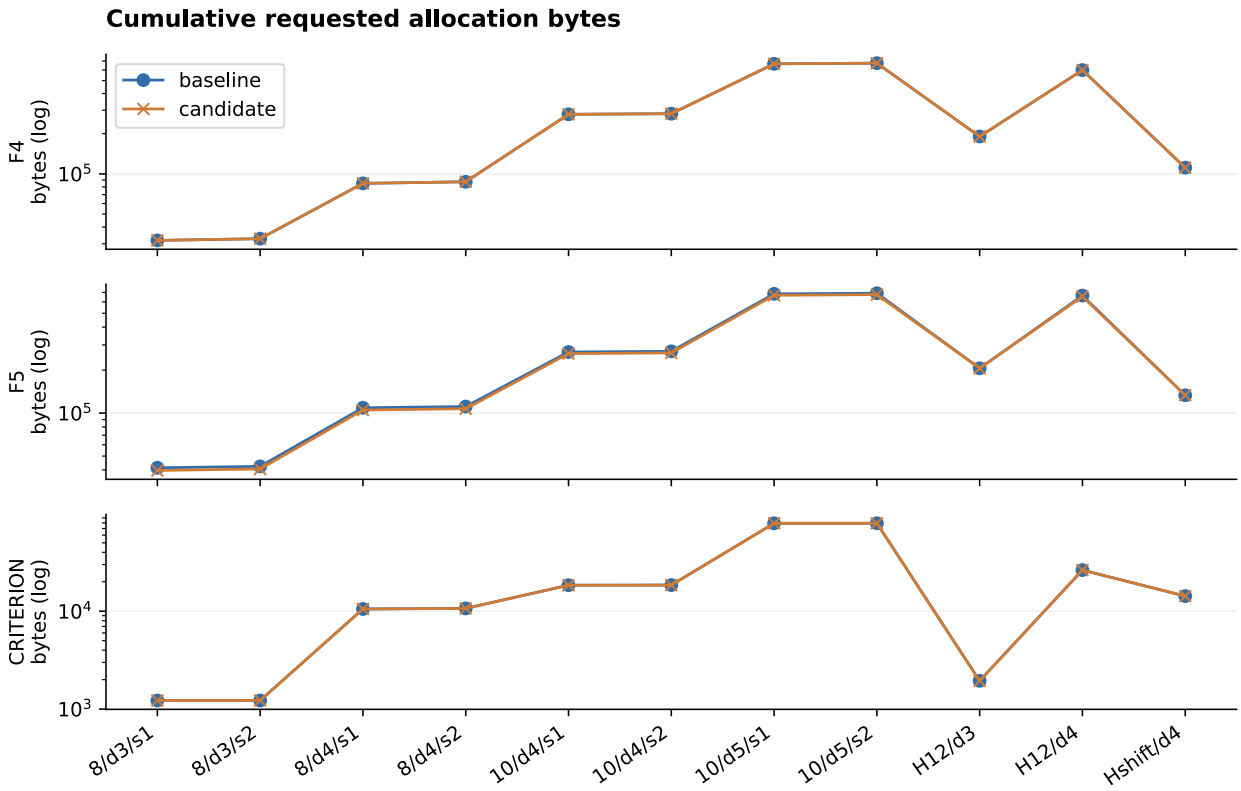
Median of 3 primary calls per case/variant; log axis. Native allocator units, not RSS.
H = known frozen regression fixture; Hshift uses bits 52–63 and 6 multiplier variables.
run-20261012T1503Z · Source: records/comparison.json.

Measured requests include packing, elimination and readback; setup, warmup, verification and JSON serialization are excluded. All primary and paired memory guards are checked, including criterion intervals. Preregistered F5/complete-call guard: True. Memory regression entries: 0. All case verdicts, including criterion regressions, remain in the native comparison.

Local validation: Release library: 3,996 passed, 0 failed, 112 ignored, exit 0, 238.05 s with loopback permission. Initial restricted run: 3,987 passed, 9 socket-permission failures, 112 ignored; retained. Focused suites: 47 Boolean Groebner, 18 F5, 21 PQ Groebner; 37 loopback checks; comparator 5 tests. All pass. Cargo example targets: 5 and 0 tests, both pass. Strict Clippy on both examples passes. Runtime ELF code/data equivalence passes.

run-20261012T1503Z; baseline 0677b1eb; candidate d56c96b9; AMD EPYC 9V74 Linux x86-64; Rust 1.99.0; Callgrind 3.24; one Rayon worker; 18 processes / 792 audited calls plus the 33-call hash control. No elapsed-time, process RSS or full-solver gain measured. Sources: native results and raw hashes, frozen protocol, report, local validation.

Cumulative requested bytes



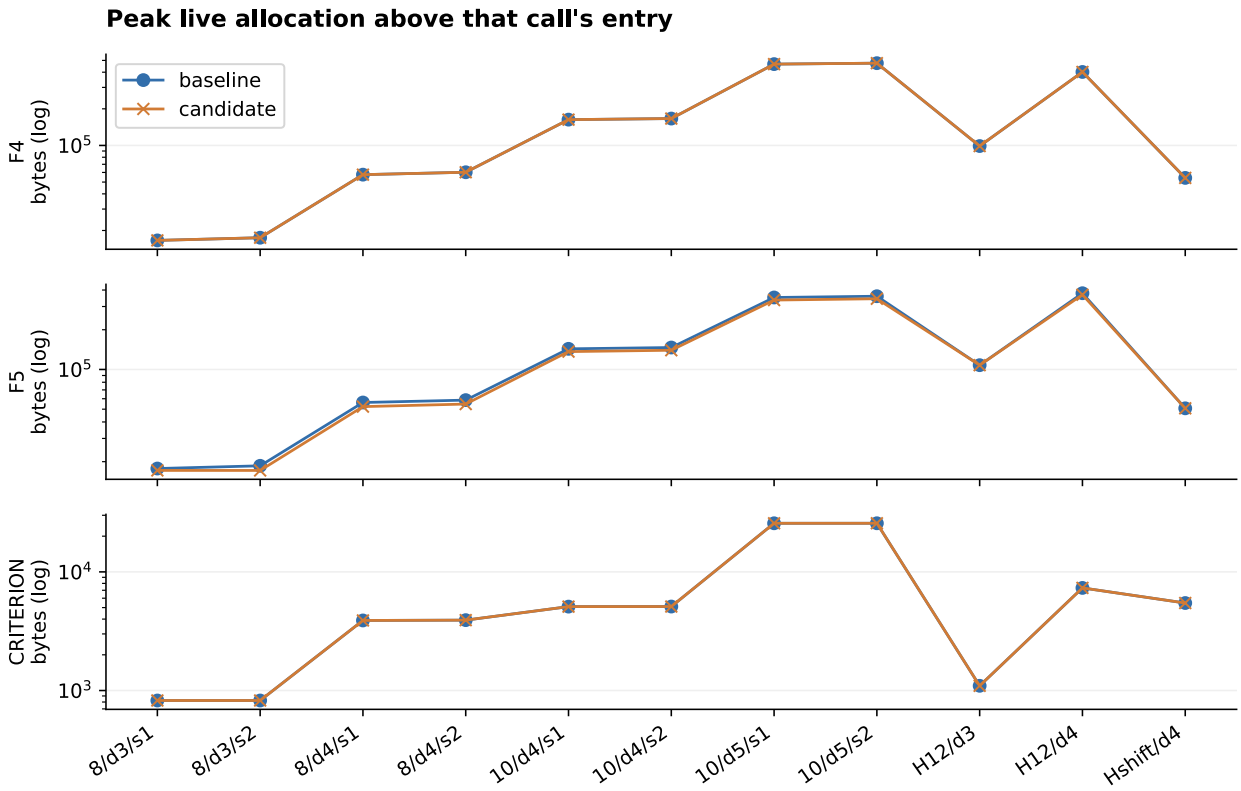
Median of 3 primary calls per case/variant; log axis. Native allocator units, not RSS.
H = known frozen regression fixture; Hshift uses bits 52–63 and 6 multiplier variables.
run-20261012T1503Z · Source: records/comparison.json.

These recorded native values include allocation and reallocation requests, not resident memory or allocation duration. Identical reduced outputs, ranks and counted word operations are required. Preregistered F5/ complete-call guard: True. Memory regression entries: 0. All case verdicts, including criterion regressions, remain in the native comparison.

Local validation: Release library: 3,996 passed, 0 failed, 112 ignored, exit 0, 238.05 s with loopback permission. Initial restricted run: 3,987 passed, 9 socket-permission failures, 112 ignored; retained. Focused suites: 47 Boolean Groebner, 18 F5, 21 PQ Groebner; 37 loopback checks; comparator 5 tests. All pass. Cargo example targets: 5 and 0 tests, both pass. Strict Clippy on both examples passes. Runtime ELF code/data equivalence passes.

run-20261012T1503Z; baseline 0677b1eb; candidate d56c96b9; AMD EPYC 9V74 Linux x86-64; Rust 1.99.0; Callgrind 3.24; one Rayon worker; 18 processes / 792 audited calls plus the 33-call hash control. No elapsed-time, process RSS or full-solver gain measured. Sources: native results and raw hashes, frozen protocol, report, local validation.

Additional peak live allocation



Median of 3 primary calls per case/variant; log axis. Native allocator units, not RSS.
H = known frozen regression fixture; Hshift uses bits 52–63 and 6 multiplier variables.
run-20261012T1503Z · Source: records/comparison.json.

The native comparator derives peak minus live-at-entry separately for each interval; inputs and retained warm outputs are outside this additional peak. Absolute peaks and entry bytes remain in raw JSONL. Preregistered F5/complete-call guard: True. Memory regression entries: 0. All case verdicts, including criterion regressions, remain in the native comparison.

Local validation: Release library: 3,996 passed, 0 failed, 112 ignored, exit 0, 238.05 s with loopback permission. Initial restricted run: 3,987 passed, 9 socket-permission failures, 112 ignored; retained. Focused suites: 47 Boolean Groebner, 18 F5, 21 PQ Groebner; 37 loopback checks; comparator 5 tests. All pass. Cargo example targets: 5 and 0 tests, both pass. Strict Clippy on both examples passes. Runtime ELF code/data equivalence passes.

run-20261012T1503Z; baseline 0677b1eb; candidate d56c96b9; AMD EPYC 9V74 Linux x86-64; Rust 1.99.0; Callgrind 3.24; one Rayon worker; 18 processes / 792 audited calls plus the 33-call hash control. No elapsed-time, process RSS or full-solver gain measured. Sources: native results and raw hashes, frozen protocol, report, local validation.